Obstructions to fixed parameters in pole certificates

Abstract

We consider the two-equation pole certificate used in the rational-source construction of the integers. No fixed rational value of its parameter e is complete on the prescribed set of prepared inputs. For either degenerate value $e = 1, -1/3$, the same prepared input $b = (204/55)^2$ gives a genus-one quartic with points over every completion of $\mathbb{Q}$ and no rational point. Exact unconditional two-descent and Cassels-pairing computations certify the two elliptic rank bounds used in the proof. An elementary squareclass argument excludes every other constant value of e. The obstruction includes zero values of the second square root, so removing a nonzero guard does not repair it. We also treat the distinct strategy of fixing a rational parameter in the first conic's parametrization. The values $\tau = \pm 1$ fail at the prepared input $b = (204/61)^2$, as certified by a third exact rank-zero calculation. The values $\tau = \pm 17$, which cancel one factor of the second twist, fail at $b = (204/101)^2$ by an exact rank-one and squareclass calculation. A rational conic through this input supplies prepared counterexamples dense in the real interval $(0, 17)$ for $\tau = \pm 17$. These results concern completeness on all prepared inputs; they do not exclude special families selected together with the comparison primes.

1 Prepared inputs and the pole equations

Definition 1.1. A rational number b is *prepared* if its reduced denominator is a square and

$$0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2},$$
$$v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2.$$

For such b, put $B = 17b$ and $V = B(B - b^2)$. The pole equations are

$$p^2 = B(1 - e)(1 + 3e), \qquad Bq^2 = (s^2 - eV)(s^2 + 3eV). \tag{1}$$

All four witnesses are rational; zero values are allowed.

These are exactly the preparation hypotheses and the zero-preparation equations stated in [1, Section 3]. Its realization theorem permits e to vary. Our question is whether a fixed value can replace that choice.

Theorem 1.2. *For every $e_0 \in \mathbb{Q}$, there is a prepared b for which (1) has no rational solution with $e = e_0$. For $e_0 = 1$ and $e_0 = -1/3$, one can take*

$$b_0 = \frac{41616}{3025} = \left(\frac{204}{55}\right)^2.$$

For either of these two values, the second equation has no rational solution even though its smooth projective quartic has points over every completion of $\mathbb{Q}$.

The rank bounds in the proof are exact computations. They use the unconditional algebraic rank bounds of PARI/GP, not an analytic-rank calculation or the Birch–Swinnerton-Dyer conjecture. The complete verifier and its exact intermediate data accompany this note.

2 Two elliptic rank certificates

For $n \in \mathbb{Q}^\times$, write

$$E_n : \quad y^2 = x(x-n)(x+3n).$$

These curves have the three rational nonidentity two-torsion points $(0,0), (n,0), (-3n,0)$.

Proposition 2.1 (Exact computation)**.** *The groups $E_{-29427}(\mathbb{Q})$ and $E_{9809}(\mathbb{Q})$ have ranks zero and one, respectively. Both have torsion subgroup $(\mathbb{Z}/2\mathbb{Z})^2$. The point*

$$P_* = (-20449, 2356926) \in E_{9809}(\mathbb{Q})$$

has infinite order.

Proof. The verifier initializes the two curves with coefficient vectors $[0, 2n, 0, -3n^2, 0]$. PARI/GP version 2.17.2 returns

n	`ellrank` output
-29427	$[0, 0, 2, []]$
9809	$[1, 1, 2, [[-20449, 2356926]]]$.

Here the first two entries are unconditional lower and upper bounds for the rational rank. The algorithm computes the two-Selmer rank, the rational two-torsion rank, and the rank of the relevant Cassels pairing; the upper bound is the first rank minus the other two [2]. The saved exact diagnostics give two-Selmer ranks $4, 5$ and pairing matrices

$$C_- = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \qquad C_+ = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Independent row reduction over $\mathbb{F}_2$ verifies that each has rank two. Since the rational two-torsion has rank two, the certified upper bounds are $4 - 2 - 2 = 0$ and $5 - 2 - 2 = 1$.

For an independent torsion check, direct finite-field counting gives

n	$\#E_n(\mathbb{F}_5)$	$\#E_n(\mathbb{F}_7)$	$\#E_n(\mathbb{F}_{11})$	$\#E_n(\mathbb{F}_{13})$
-29427	4	8	8	12
9809	8	8	16	12.

All these primes are good. The injection on prime-to-p torsion under good reduction, using the different primes in the table, bounds the rational torsion order by four in each case. The known two-torsion attains this bound. Direct substitution verifies P_*, and its nonzero ordinate shows that it is not torsion. This also independently supplies the lower rank bound one for E_{9809}. □

Lemma 2.2. *For a nonsingular curve $y^2 = x(x^2 + ax + c)$ with $c \neq 0$, the map*

$$\alpha : E(\mathbb{Q}) \longrightarrow \mathbb{Q}^\times/\mathbb{Q}^{\times 2}, \qquad \alpha(O) = 1, \quad \alpha((0,0)) = c, \quad \alpha((x,y)) = x \quad (x \neq 0)$$

is a group homomorphism. It kills $2E(\mathbb{Q})$. In particular, if E has rank r and full rational two-torsion, its image has size at most 2^{r+2}.

Proof. Intersect a nonvertical line $y = ux + v$ with the curve. The product of its three intersection abscissas, counted with multiplicity, is v^2. Since reflection in the x-axis does not change an abscissa, this proves the homomorphism identity when $v \neq 0$. If $v = 0$, the two nonzero intersection abscissas have product c, which gives the identity when the third point is $(0, 0)$. Addition by $(0, 0)$ sends a nonzero abscissa x to c/x; vertical lines and two-torsion doubling give the remaining cases. The target has exponent two, so $2E(\mathbb{Q})$ is killed. Finally the Mordell–Weil decomposition gives $|E(\mathbb{Q})/2E(\mathbb{Q})| = 2^r|E(\mathbb{Q})[2]| = 2^{r+2}$. □

Corollary 2.3. *Neither $E_{-29427}(\mathbb{Q})$ nor $E_{9809}(\mathbb{Q})$ has a nonzero abscissa in the rational squareclass of* 17.

Proof. The first curve has only its four torsion points. Its nonzero abscissas are -29427 and $88281 = 9 \cdot 17 \cdot 577$; neither has squareclass 17.

On E_{9809}, the two-torsion points give squareclasses -3 and 9809, while P_* gives -1, because $20449 = 143^2$. These three classes are independent: $9809 = 17 \cdot 577$ and the prime 577 distinguishes the third generator from the first two. They therefore supply eight classes. By Lemma 2.2 and rank one, these exhaust the image:

$$\alpha(E_{9809}(\mathbb{Q})) = \{\pm 1,\ \pm 3,\ \pm 9809,\ \pm 29427\}.$$

The class 17 is absent. □

3 The two degenerate specializations

Put $M = 55$, $t = 204/M$, and $d = M^2 - 2448 = 577$. Then

$$b_0 = t^2, \qquad B = 17t^2, \qquad 17 - t^2 = \frac{17d}{M^2}, \qquad V = \frac{289t^4 d}{M^2}.$$

For $e = 1$, the invertible rational substitutions

$$s = \frac{17t^2}{M}u, \qquad q = \frac{17t^3}{M^2}v$$

turn the second pole equation into

$$C_+ : \quad v^2 = 17(u^2 - d)(u^2 + 3d). \tag{2}$$

For $e = -1/3$, the substitutions

$$s = \frac{17t^2}{3M}u, \qquad q = \frac{17t^3}{9M^2}v$$

give

$$C_- : \quad v^2 = 17(u^2 + 3d)(u^2 - 9d). \tag{3}$$

The verifier checks these identities coefficient by coefficient.

In either equation $u = 0$ makes the right side negative. Every rational affine point would therefore have $u \neq 0$. The map

$$(x, y) = (17u^2, 17uv)$$

sends C_+ to $E_{17d} = E_{9809}$ and C_- to $E_{-51d} = E_{-29427}$. Its nonzero abscissa has squareclass 17, contradicting Corollary 2.3. This proves that both quartics have no rational affine point. Their points at infinity would require 17 to be a rational square, so their smooth projective completions have no rational point either.

Proposition 3.1. *Both C_+ and C_- have points over every completion of $\mathbb{Q}$.*

Proof. Let $g_\pm(u)$ denote the corresponding right side. The following table gives rational values of u for which $g_\pm(u)$ is a nonzero square in the indicated local field. Each entry records u, the valuation, and the residue of the unit after removing that valuation. The unit is taken modulo 8 at 2 and modulo p at an odd prime.

p	$C_+ : (u, v_p(g_+), \text{unit})$	$C_- : (u, v_p(g_-), \text{unit})$
2	$(23, 6, 1)$	$(21, 6, 1)$
3	$(1, 2, 1)$	$(12, 4, 1)$
17	$(4, 2, 13)$	$(22, 2, 9)$
577	$(1, 0, 17)$	$(1, 0, 17)$.

The units $13, 9$ are squares modulo 17, and 17 is a square modulo 577. The standard local square criterion proves these four local assertions. At the real place, $u = 100$ makes both right sides positive.

The quartic discriminants are

$$\operatorname{disc}(g_+) = -2^{12}3\,17^6 d^6, \qquad \operatorname{disc}(g_-) = -2^{12}3^7\,17^6 d^6.$$

Thus the smooth projective genus-one curves have good reduction outside $2, 3, 17, 577$. At any other prime, the Hasse bound gives a point on the reduction, and smooth lifting gives a local point. This proves the proposition. □

4 Exclusion of all other constant values

The number $b_0 = (204/55)^2$ is prepared: its reduced denominator is square, it lies in $(0, 17)$, and its valuations at $2, 3, 17$ are $4, 2, 2$. Being a rational square, it satisfies all three local square conditions.

A second prepared number is

$$b_1 = 409\left(\frac{204}{1001}\right)^2 = \frac{17020944}{1002001}.$$

It is less than 17 because $17020944 < 17034017$. Its denominator is square and its valuations at $2, 3, 17$ are again $4, 2, 2$. The number 409 is 1 modulo $8, 3, 17$, so it is a square in all three local fields.

Suppose $e_0 \neq 1, -1/3$ made the first pole equation soluble for both inputs. Its factor $(1 - e_0)(1 + 3e_0)$ is nonzero; division of the two equations would make b_1/b_0 a rational square. But

$$\frac{b_1}{b_0} = 409\left(\frac{55}{1001}\right)^2$$

is not a square. Thus at least one of the two prepared inputs already fails the first equation. The two excluded degenerate values fail at b_0 by Section 3. This completes the proof of Theorem 1.2.

The theorem concerns a constant e. It does not exclude a nonconstant function of b, a different relation between the pole parameters, or a different pole predicate. Those proposals require their own completeness arguments.

5 A fixed parameter in the conic chart

For a nonzero rational pole input b, write $B = 17b$ and

$$e = \frac{\tau^2 - B}{\tau^2 + 3B}, \qquad p = \frac{-4B\tau}{\tau^2 + 3B}.$$

Fixing τ is different from fixing e, since e still varies with b. The following exact example rules out the choice $\tau = 1$, as well as $\tau = -1$, as a complete specialization on all prepared pole inputs.

Proposition 5.1. *The rational number*

$$b = \left(\frac{204}{61}\right)^2 = \frac{41616}{3721}$$

has square reduced denominator, satisfies $0 < b < 17$*, is a square in* $\mathbf{Q}_2, \mathbf{Q}_3, \mathbf{Q}_{17}$*, and has respective valuations* $4, 2, 2$*. For* $\tau = 1$ *or* $\tau = -1$*, the equation*

$$Bq^2 = (s^2 - Be(B - b^2))(s^2 + 3Be(B - b^2))$$

has no solution $(s, q) \in \mathbf{Q}^2$*.*

Proof. The preparation assertions follow directly from $b = (204/61)^2$. For either choice of τ,

$$e = -\frac{703751}{2126137}.$$

Put $M = Be(B - b^2)$ and $N = BM$. Direct rational arithmetic gives

$$N = nk^2, \qquad n = -32380801574194567, \qquad k = \frac{144324288}{29438154846217}.$$

A hypothetical solution with $s \neq 0$ would give the point

$$x = \frac{B}{k^2}s^2, \qquad y = \frac{B^2}{k^3}sq$$

on

$$E_n : \quad y^2 = x(x - n)(x + 3n).$$

Its nonzero abscissa would belong to the squareclass 17.

Exact two-descent gives $\operatorname{rank} E_n(\mathbf{Q}) = 0$. Its saved diagnostic has two-Selmer rank two, equal to the dimension of the full rational two-torsion. Thus no Cassels-pairing refinement is required for this rank-zero certificate. The accompanying reproducible PARI calculation returns `ellrank` equal to `[0,0,0,[]]`; it uses no analytic-rank or Birch–Swinnerton-Dyer hypothesis. Independent counts at the good primes $5, 7, 11, 23, 29, 31$ are respectively

$$4, 8, 8, 32, 24, 24.$$

Their greatest common divisor is four. Together with the visible full two-torsion this proves

$$E_n(\mathbf{Q}) = \{O, (0, 0), (n, 0), (-3n, 0)\}.$$

Neither nonzero torsion abscissa has squareclass 17: n is negative, and $-3n/17$ is not a rational square. This excludes $s \neq 0$. If $s = 0$, the original equation becomes $Bq^2 = -3M^2 < 0$, which is impossible over $\mathbf{R}$. This proves the assertion. □

5.1 A parameter which cancels one twist factor

For $\tau = 17$ or $\tau = -17$ the first conic gives

$$e = \frac{17 - b}{17 + 3b}.$$

The elliptic parameter governing the second equation is

$$N = B^2e(B - b^2) = b(17 + 3b)\left(\frac{17b(17 - b)}{17 + 3b}\right)^2.$$

Thus its squareclass no longer contains the factor $17 - b$. This simplification does not ensure completeness.

Proposition 5.2. *For $b = (204/101)^2$, neither choice $\tau = 17, -17$ admits a rational solution of the second pole equation. The input is prepared.*

Proof. The preparation follows exactly as above: its valuations at $2, 3, 17$ are $4, 2, 2$, and it is a positive rational square less than 17. Direct calculation gives

$$e = \frac{7753}{17545}, \qquad N = 2465k^2, \qquad k = \frac{1118946204864}{165976339595}.$$

The same map $x = (B/k^2)s^2$, $y = (B^2/k^3)sq$ sends a solution with $s \neq 0$ to a point of $E_{2465}(\mathbb{Q})$ with abscissa in squareclass 17.

Exact two-descent gives two-Selmer rank three and hence rational rank at most one. The good-prime counts at $7, 11, 13, 19, 23, 31$ are $8, 8, 12, 16, 32, 24$, respectively, so the rational torsion has order four. The point

$$(77571, 22248954) \in E_{2465}(\mathbb{Q})$$

has nonzero ordinate and therefore infinite order. This proves rank one independently of any choice of a Mordell–Weil basis. Its abscissa has squareclass 51, since $77571 = 51 \cdot 39^2$. Together with the torsion squareclasses -3 and 2465, it gives the eight classes

$$\{1, -3, 2465, -7395, 51, -17, 435, -145\}.$$

These are distinct, as is verified by their prime factorizations. Lemma 2.2 bounds the entire image by eight classes, so this list is exhaustive without a saturation computation. The required class 17 is absent. Finally $s = 0$ would again give $Bq^2 = -3M^2 < 0$. This proves the assertion. □

5.2 Prepared counterexamples in every real interval

Theorem 5.3. *The prepared rational numbers for which neither choice $\tau = 17, -17$ admits a rational solution of the second pole equation are dense in the real interval $(0, 17)$. In particular, for every $\epsilon > 0$ there are infinitely many such numbers in $(0, \epsilon)$. They can all be chosen to be rational squares with valuations $v_2(b) = 4$, $v_3(b) = 2$, and $v_{17}(b) = 2$.*

Proof. Consider the nonsingular rational conic

$$17 + 3u^2 = 2465v^2. \tag{4}$$

It contains the point

$$(u_0, v_0) = (204/101, 11/101).$$

For a rational solution with $0 < u^2 < 17$, put $b = u^2$ and $e = (17-b)/(17+3b)$. The parameter of the second elliptic curve is

$$N = B^2e(B - b^2) = \frac{289u^6(17 - u^2)^2}{2465v^2} = 2465\left(\frac{17u^3(17 - u^2)}{2465v}\right)^2.$$

Here $u, v, 17-u^2$ are nonzero. The rational map used in Proposition 5.2 would therefore send a pole solution with $s \neq 0$ to a point of $E_{2465}(\mathbb{Q})$ with nonzero abscissa in squareclass $B = 17u^2$, namely class 17. The exhaustive squareclass calculation in that proposition excludes this class. The case $s = 0$ gives $Bq^2 = -3M^2 < 0$, as before. Thus every such point on the conic supplies an obstruction.

It remains to impose preparation and an arbitrary open real interval simultaneously. A parametrization obtained by drawing the line of slope t through (u_0, v_0) is

$$\begin{aligned} u(t) &= \frac{-u_0(3+2465t^2)+4930v_0t}{3-2465t^2}, \\ v(t) &= \frac{v_0(3+2465t^2)-6u_0t}{3-2465t^2}. \end{aligned} \tag{5}$$

At the rational tangent parameter $t_0 = 3u_0/(2465v_0)$ it gives (u_0, v_0); its denominator is nonzero there. Consequently, for each of $p = 2, 3, 17$, there is an open p-adic neighborhood of t_0 on which

$$v_2(u) = 2, \qquad v_3(u) = 1, \qquad v_{17}(u) = 1$$

holds at the respective place.

Fix any nonempty open interval $I \subset (0, 17)$ and a point $b_* \in I$. The conic contains the real point $(\sqrt{b_*}, \sqrt{(17+3b_*)/2465})$. It is a finite point of the parametrization: the vertical line through (u_0, v_0) meets the positive branch only at the base point, which is represented by t_0. Choose a sufficiently small open real interval in parameter space on which $u(t)^2 \in I$ and all denominators are nonzero. Weak approximation for $\mathbb{Q}$ gives infinitely many rational parameters in this interval and in the three prescribed p-adic neighborhoods. The nonconstant rational function $u(t)^2$ has finite fibers, so their values $b = u(t)^2$ form an infinite set.

Each b is a nonzero rational square, hence has square reduced denominator and is a square in all three local fields. Its valuations are exactly $4, 2, 2$, and its real value belongs to the interval $I \subset (0, 17)$. It is therefore prepared. The obstruction proved in the first paragraph applies to every one of these inputs. □

These propositions concern all prepared inputs. They do not decide whether a fixed conic parameter can be made complete on a more restricted family chosen simultaneously with the comparison primes.

Reproducible verification

Run `check_fixed_e_obstruction.py` to reproduce `fixed-e-obstruction-certificate.json` and the descent transcript `fixed-e-pari-descent.txt`. The script independently checks preparation, substitutions, torsion, squareclasses, local witnesses, and pairing-matrix ranks using exact integer and rational arithmetic. The two algebraic rank calculations use PARI; no bounded search is used as a nonexistence certificate.

Run `check_fixed_tau_obstruction.py` to reproduce `fixed-tau-one-obstruction.json` and `fixed-tau-one-pari-descent.txt`. It verifies the conic parameter, the rational map to the rank-zero elliptic curve, the complete torsion obstruction, and all preparation conditions. Its saved two-Selmer rank independently supplies the stated upper bound.

The command

```
check_fixed_tau17_obstruction.py
```

reproduces the two files

```
fixed-tau17-obstruction.json
fixed-tau17-pari-descent.txt.
```

It checks the factor cancellation, the rational scaling, the explicit nontorsion point, and all eight squareclasses. Their exhaustive nature follows from the proved rank bound, not from assuming that the displayed point generates the free part.

The command

`check_small_fixed_tau17_obstruction.py`

reproduces `small-fixed-tau17-obstruction.json`. It verifies the conic parametrization and twist identity, the same exact rank and squareclass obstruction, and an explicit prepared rational square $0 < b < 10^{-80}$. The infinitely many counterexamples in Theorem 5.3 follow from its weak-approximation proof.

References

[1] *Arithmetic dependencies of the construction with 29 variables*, project paper, version 8, 8 November 2026.

[2] The PARI Group, *PARI/GP: Elliptic curves*, documentation of `ellrank`, including its unconditional two-descent and Cassels-pairing bounds, `https://pari.math.u-bordeaux.fr/dochtml/html/Elliptic_curves.html#ellrank`.